MOUNT KENYA UNIVERSITY

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ADMISSION NUMBER:BSCCS/2024/73613

COURSE: COMPUTER SCIENCE

UNIT NAME: ADVANCE WEB DESIGN AND DEVELOPMENT

LECTURER NAME: MICHAEL NYORO

SEMESTER: YEAR 3 SEM 2

PROJECT TITLE:

TECHNOLOGIES: PHP + MYSQL

GITHUB LINK:
https://github.com/MrsNjenga/my_project.git

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WEEK ONE: LOCAL ENVIRONMENT SETUP

Title: installation and testing of local development environment

N/B: I had already set up the environment earlier

WEEK TWO: WIREFRAMES AND GUI DESIGN

Title:User interface planning and system design

Fig1: login wireframe

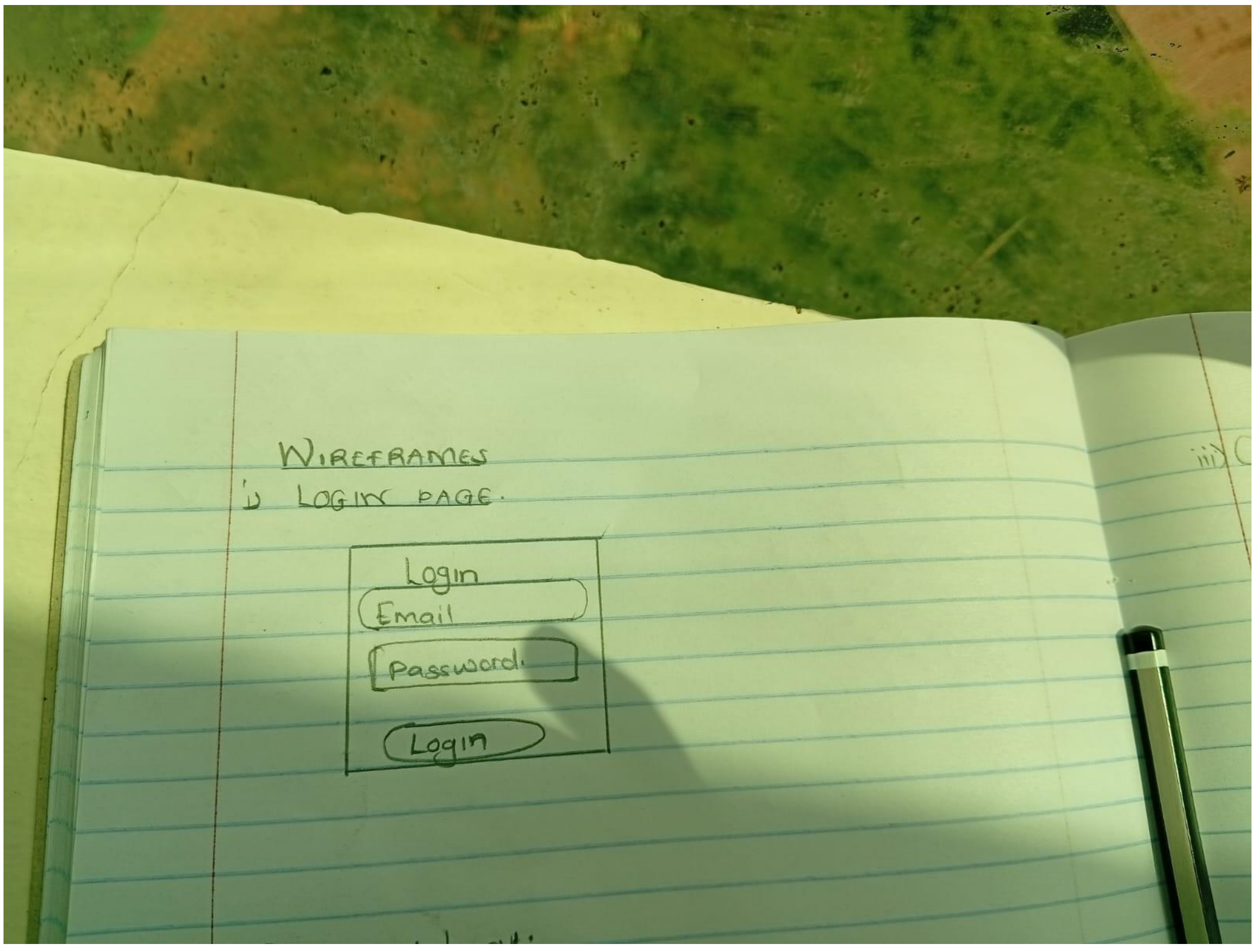


Fig2:Dashboard layout

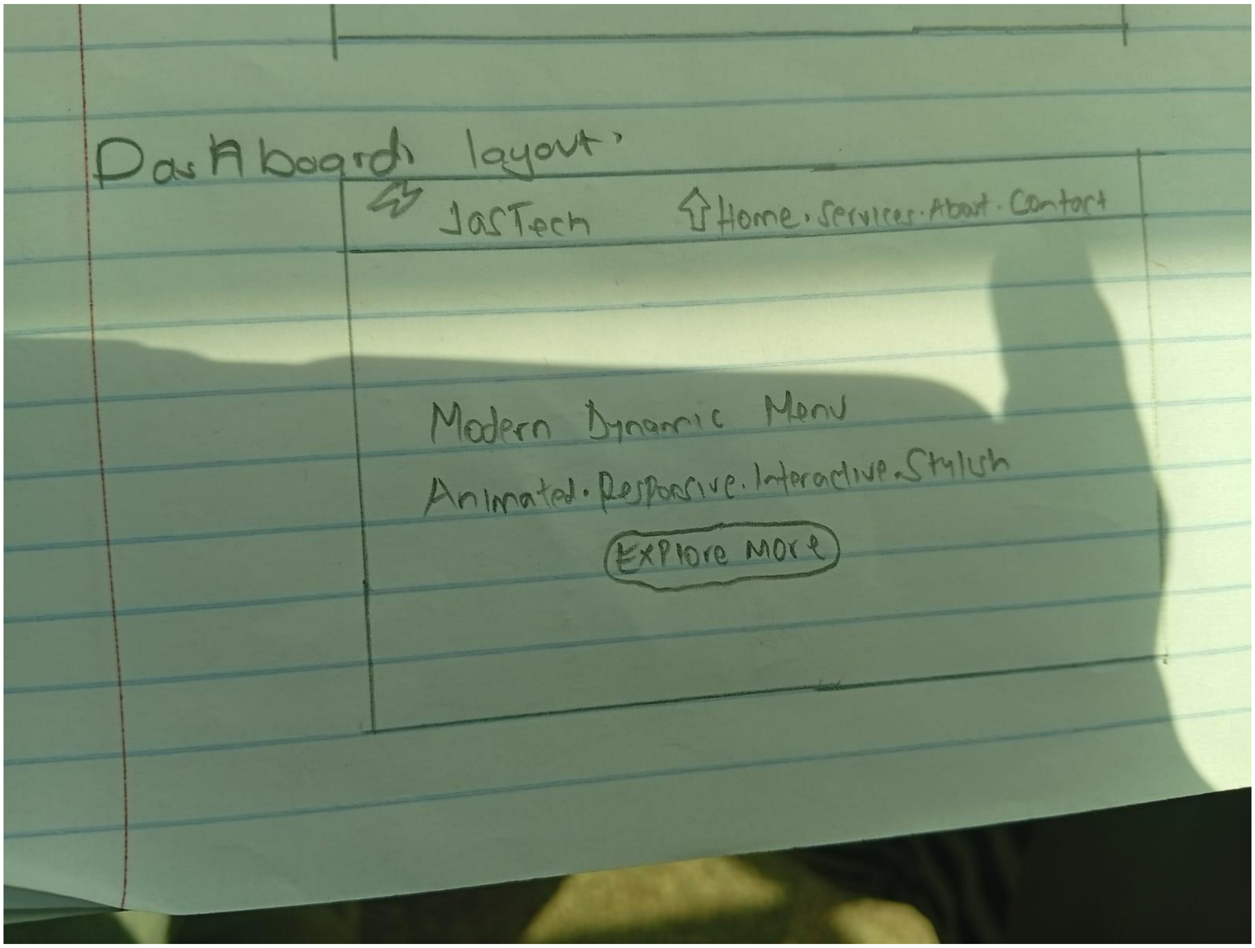
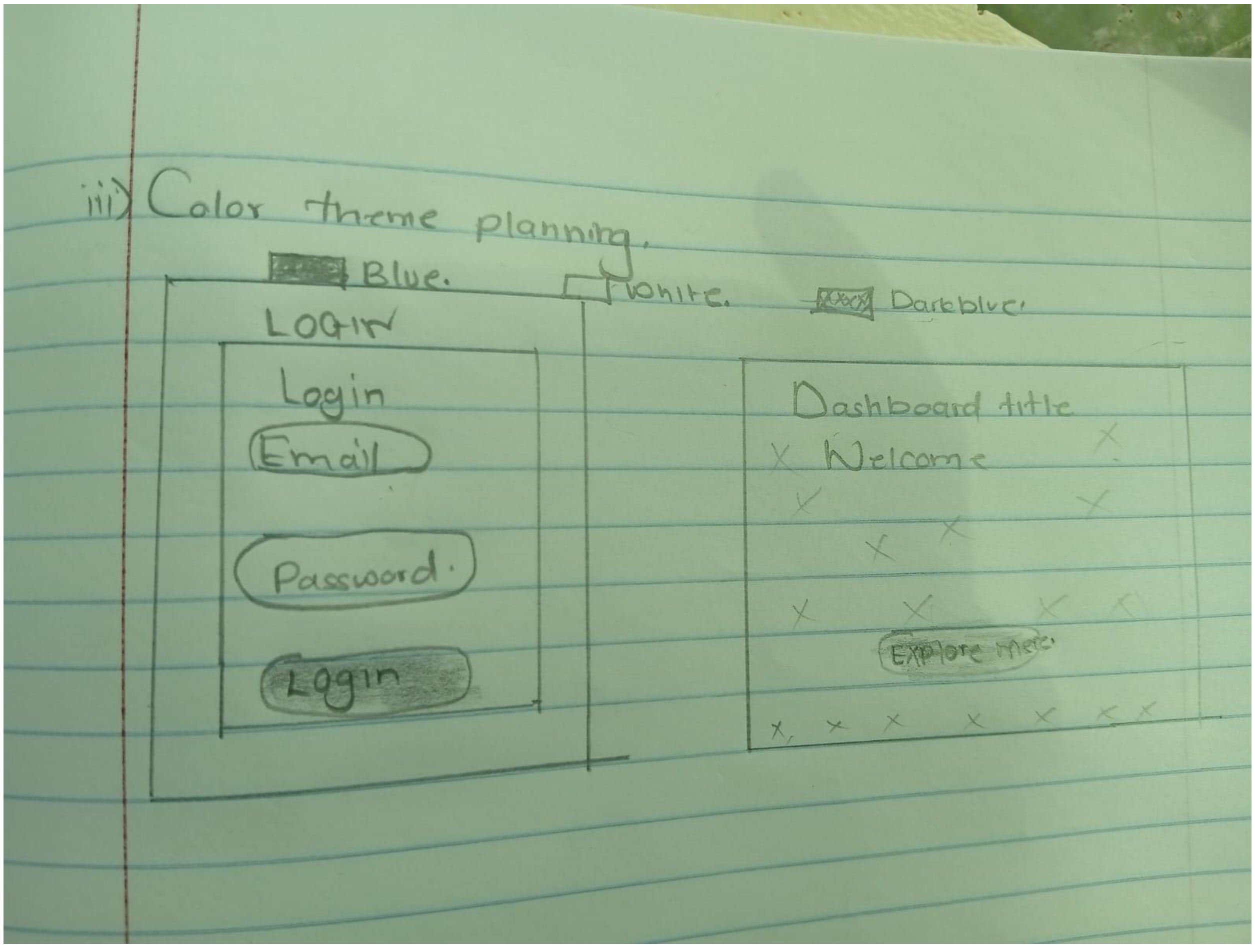


Fig3: Color theme planning



WEEK THREE: JAVASCRIPT AND PHP BASICS

Title:Frontend interaction and Backend

Evidences:

Fig1: Javascript form validation preventing empty submissions

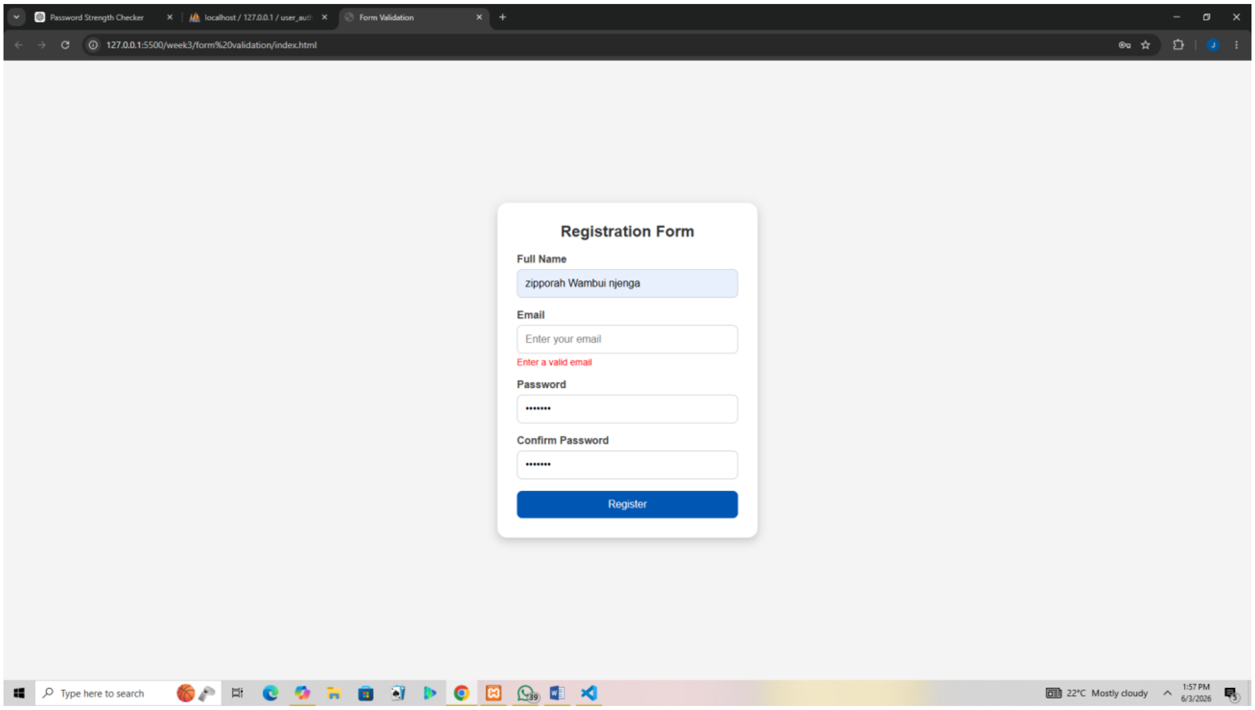


Fig2: Password strength checker with dynamic feedback

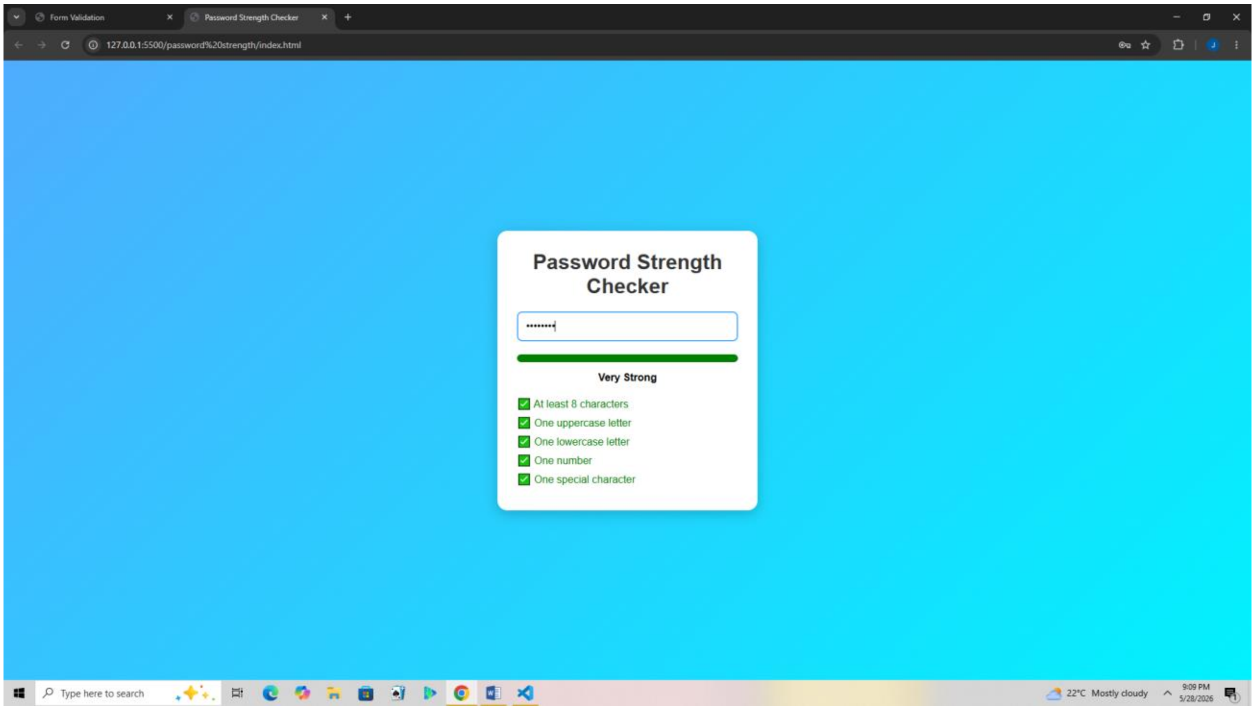


Fig3:Email validation script ensuring correct format

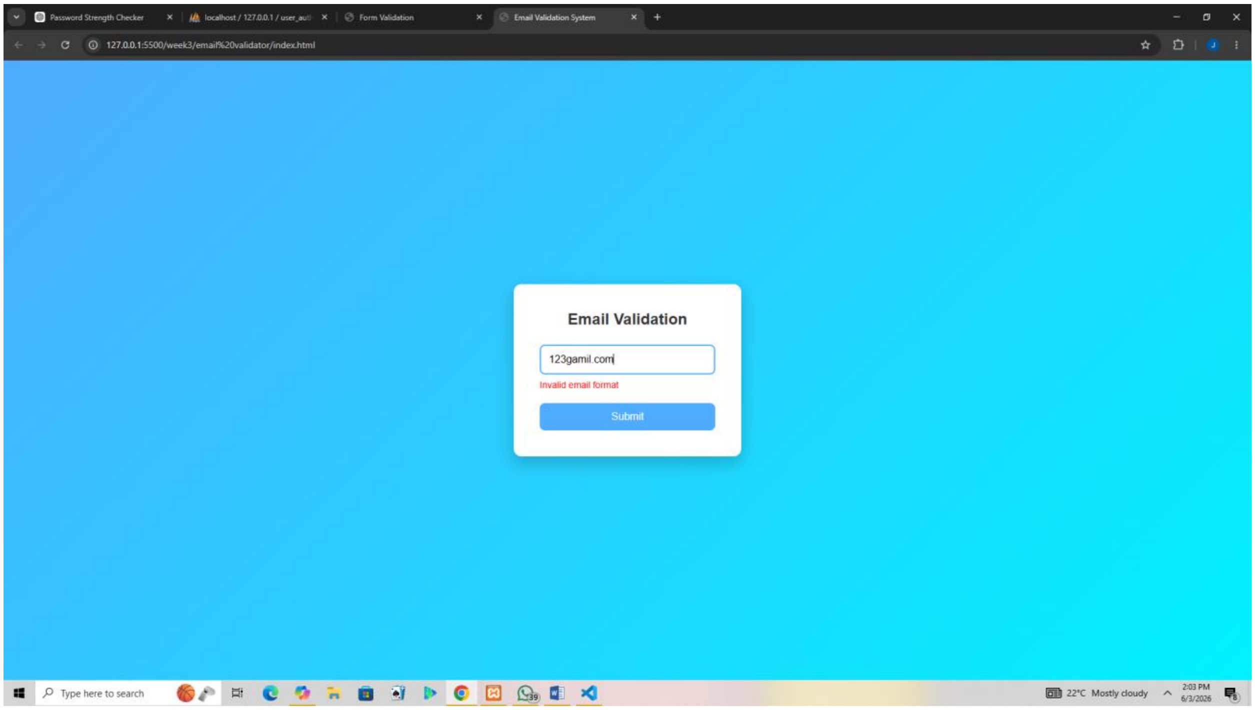


Fig4: login form validation with error message

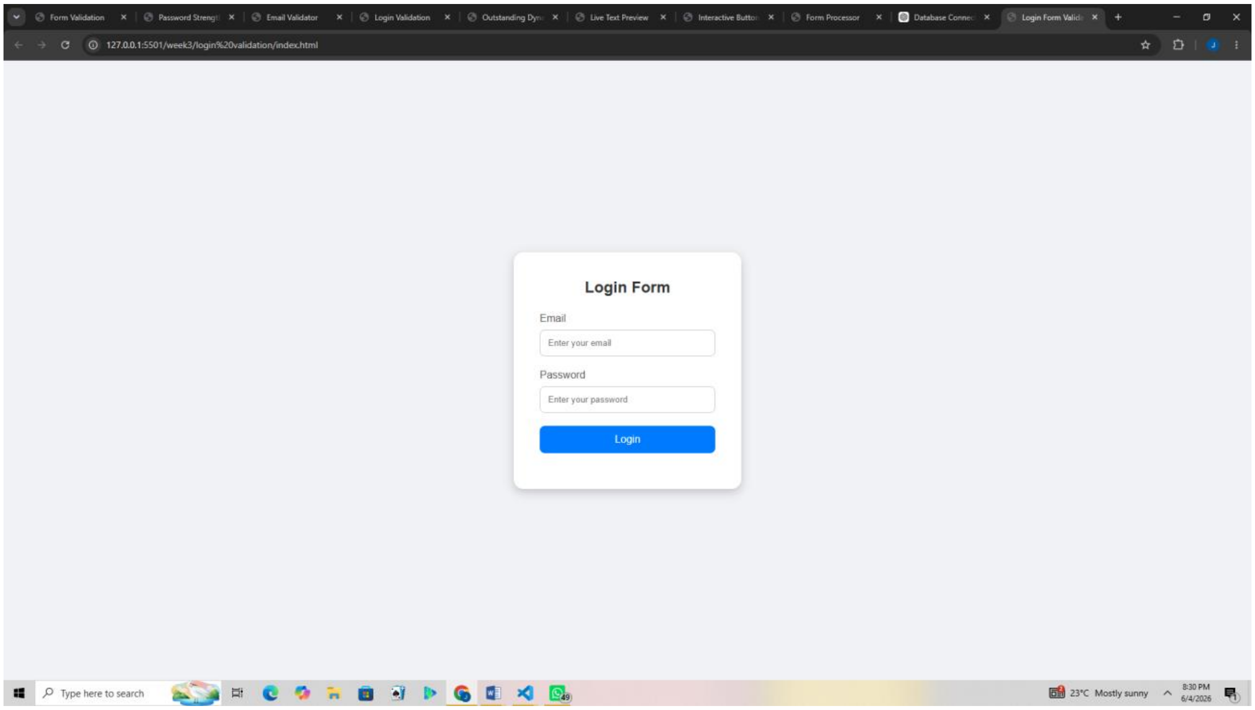


Fig5: Dynamic menu created with javascript

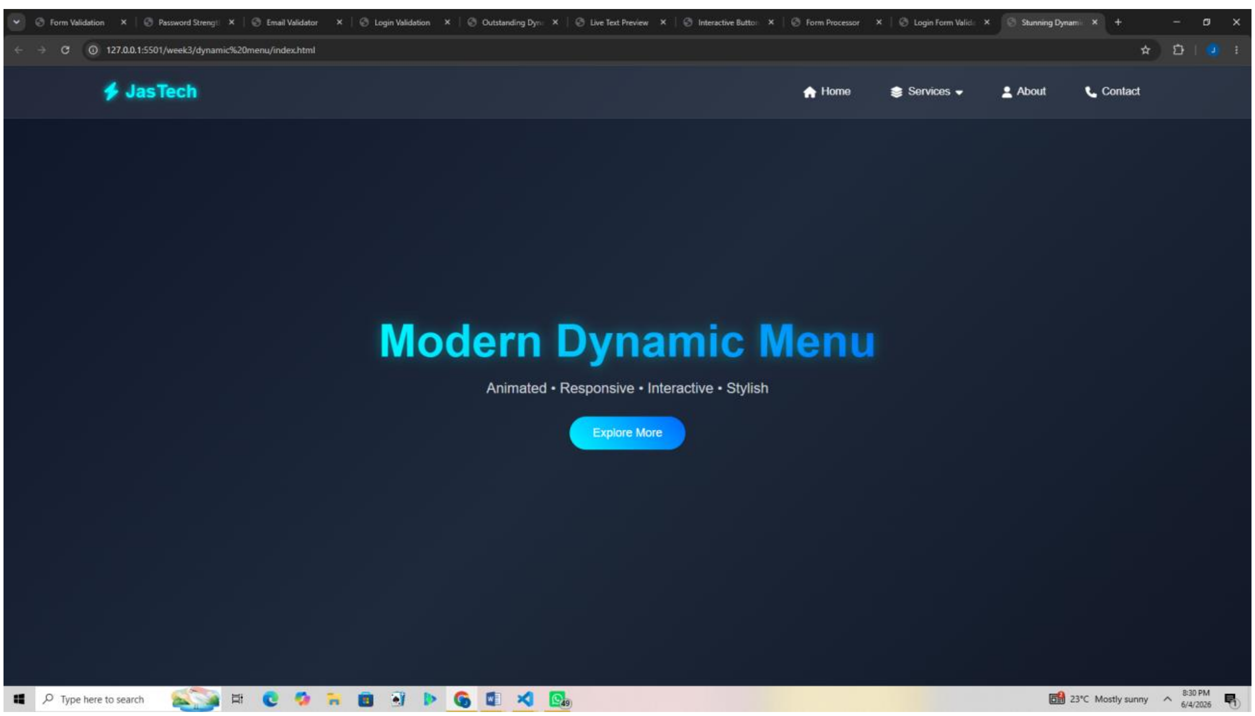


Fig6: live text preview system updating content in real time

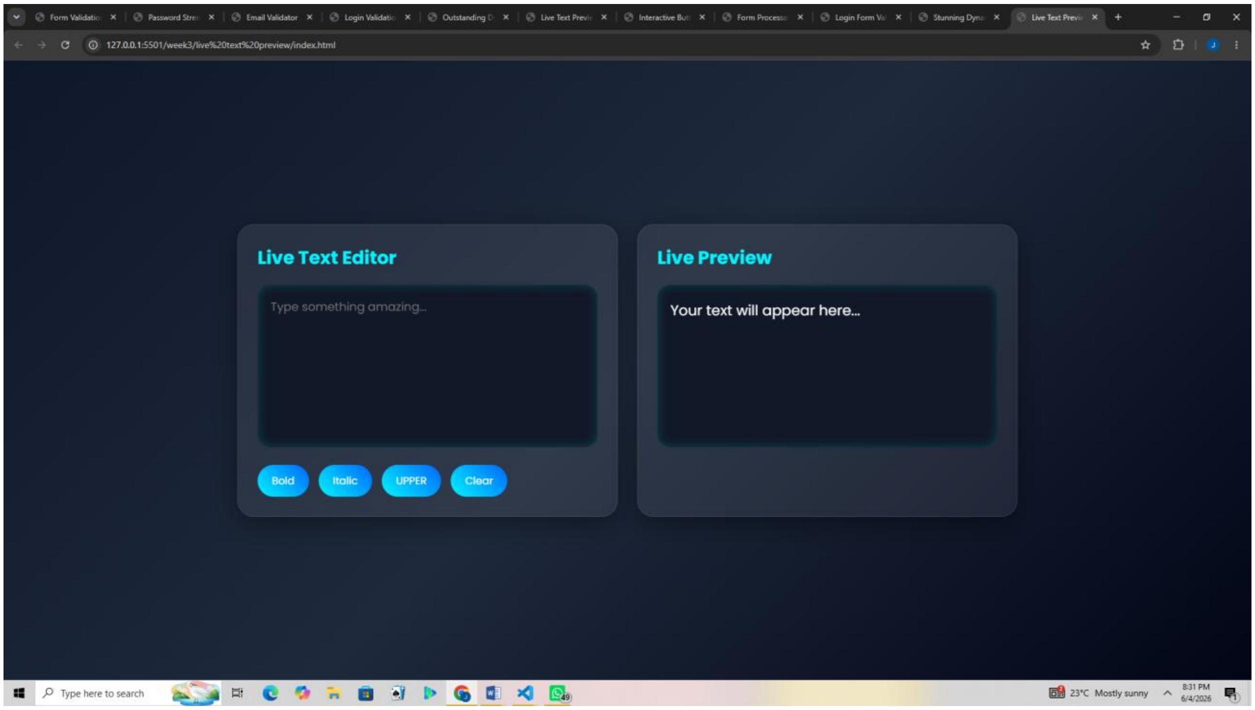
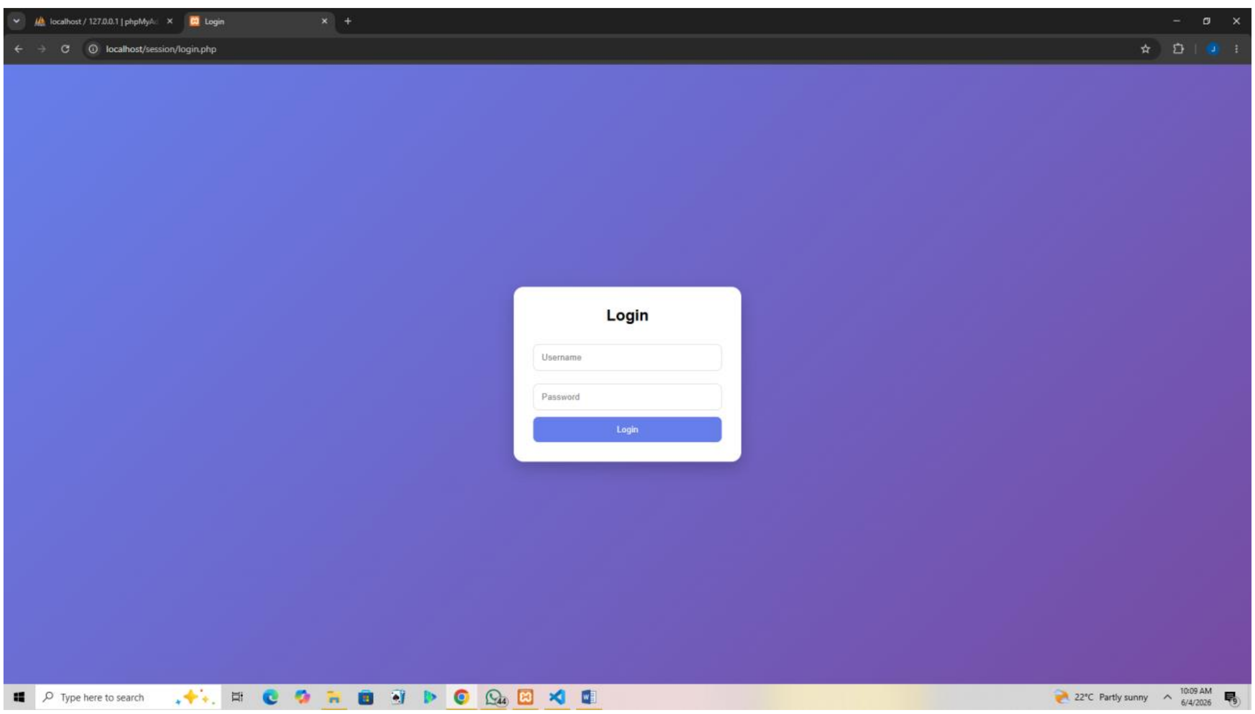


Fig7: index.php file displaying welcome message



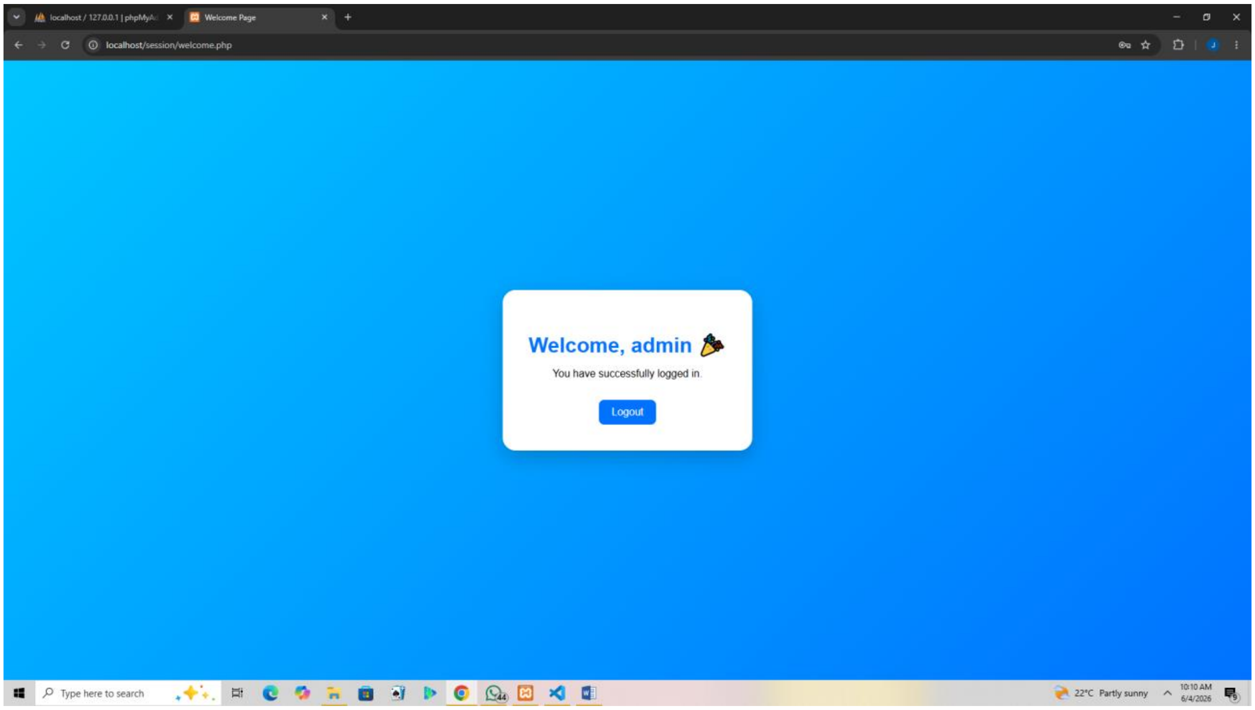


Fig8:interactive buttons responding to script

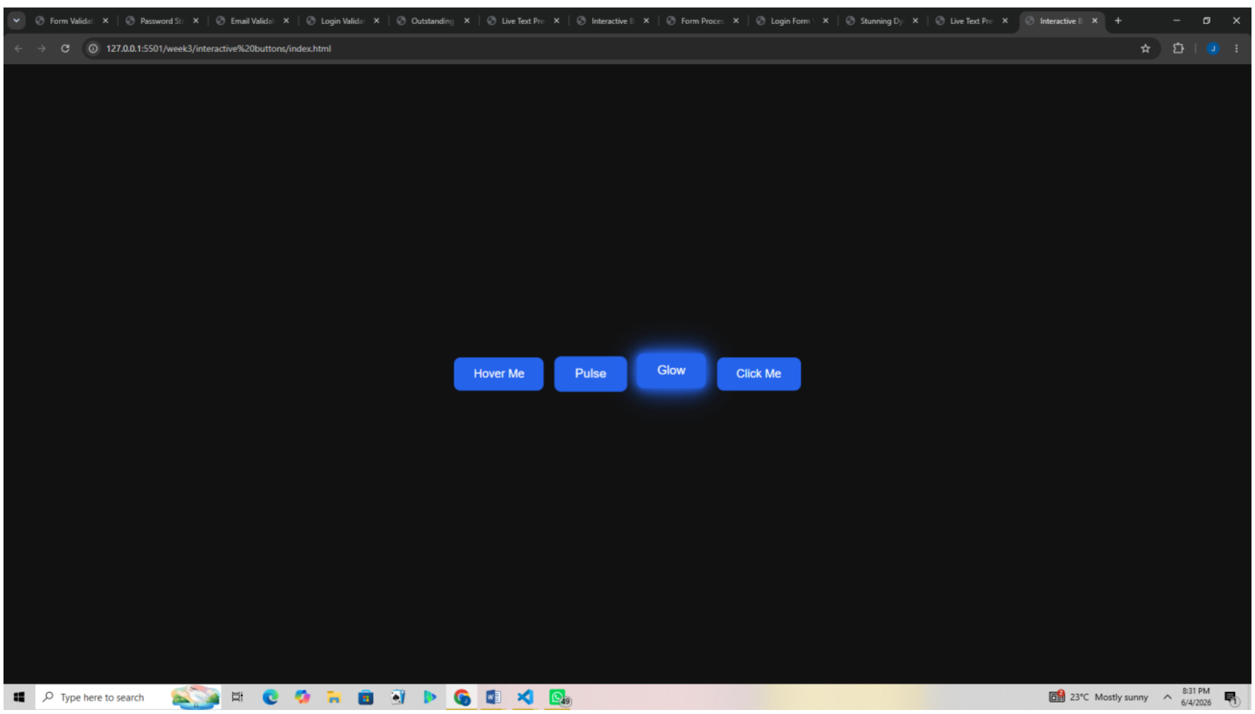
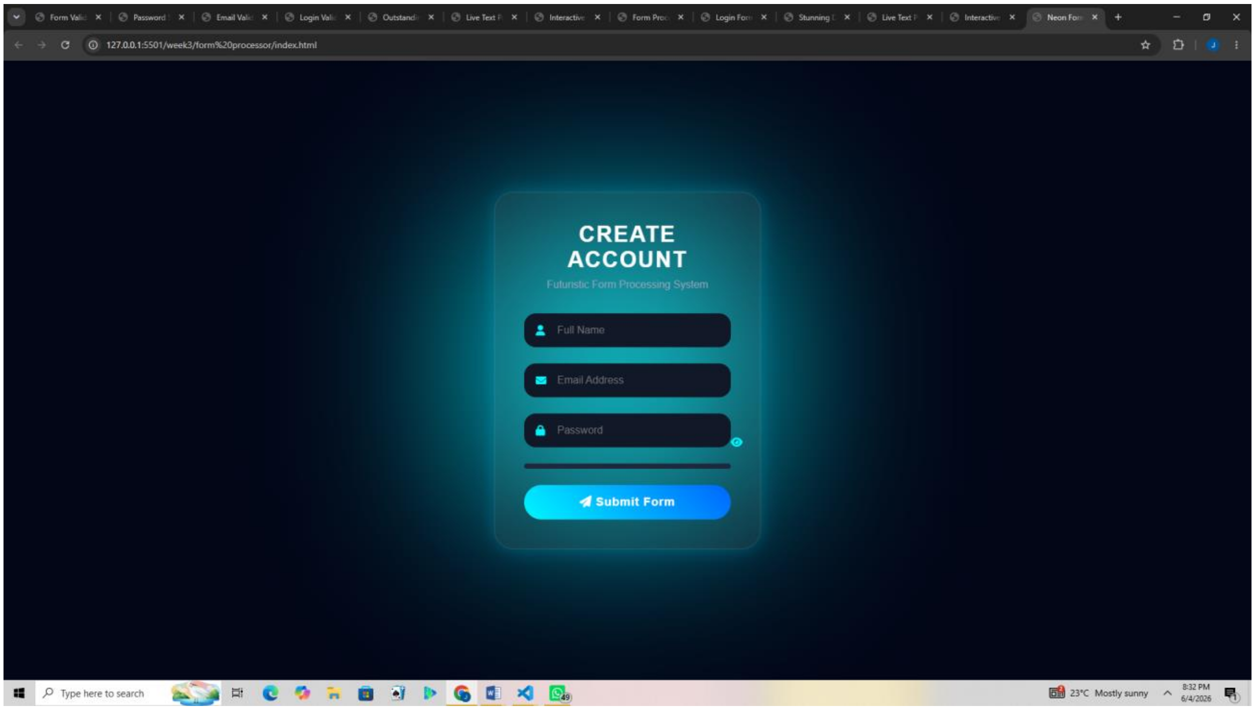


Fig9: simple form processor



REFLECTION

This week I advanced from static pages to interactive systems. JavaScript tasks (Fig 1–7) improved user experience through validation, dynamic menus, and live previews. PHP basics (Fig 8–9) introduced server-side scripting and form handling, while Fig 10 confirmed successful database connectivity. These tasks demonstrated how frontend scripts and backend code integrate to build dynamic applications. The main challenge was debugging validation scripts and PHP syntax errors, which taught me careful attention to detail and improved my problem-solving skills:

WEEK 4:SERVER-SIDE COMPONENTS & BACKEND

Fig 1: Form submission system displaying dynamic welcome message

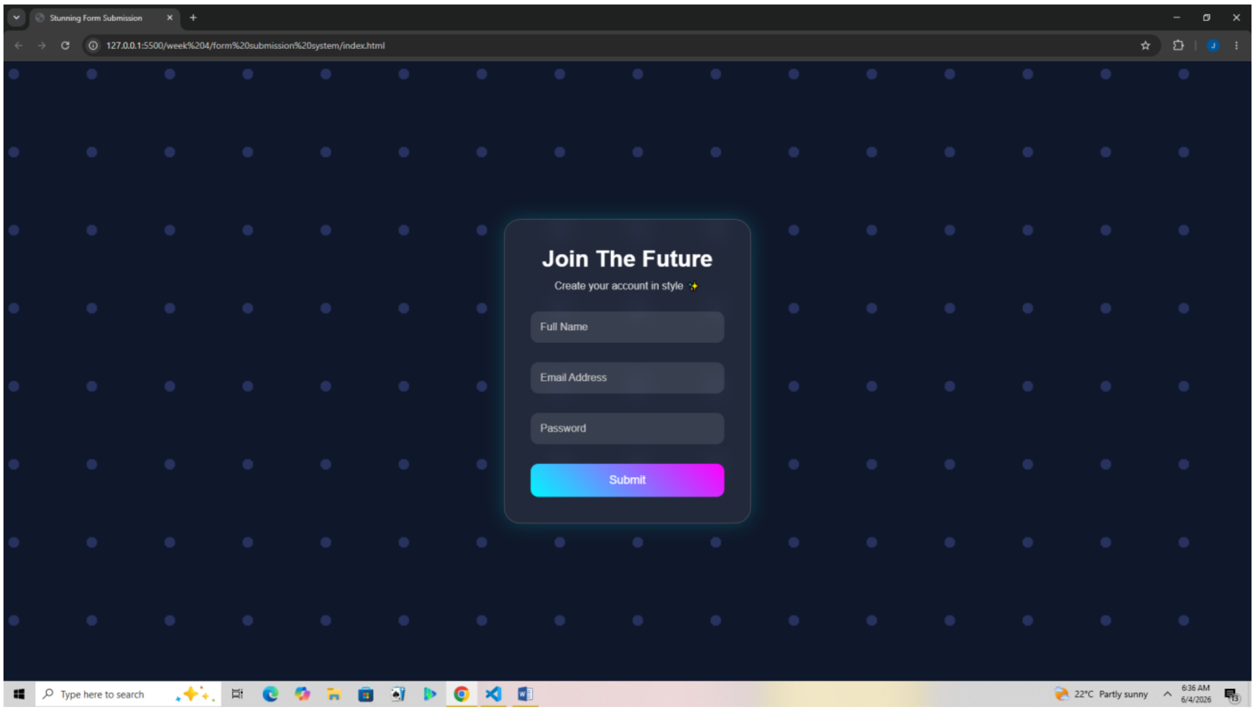


Fig 3: Registration form integrated with POST method

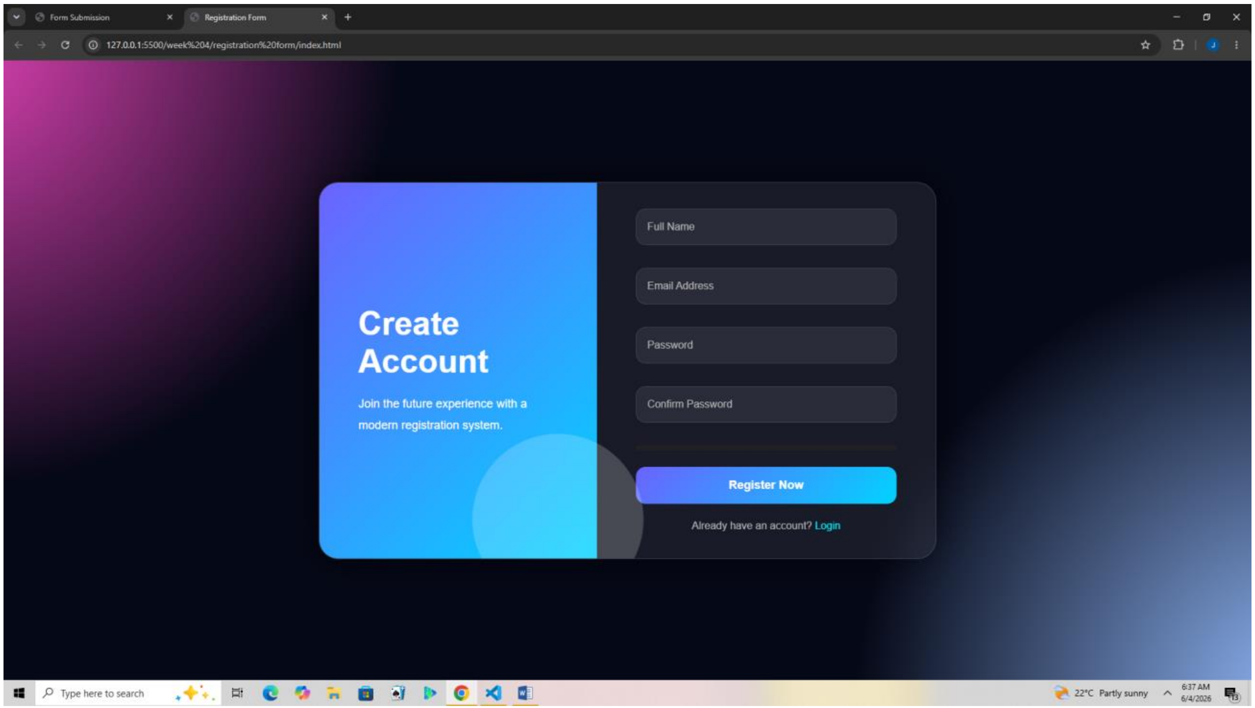


Fig 4: Login form with username/password validation

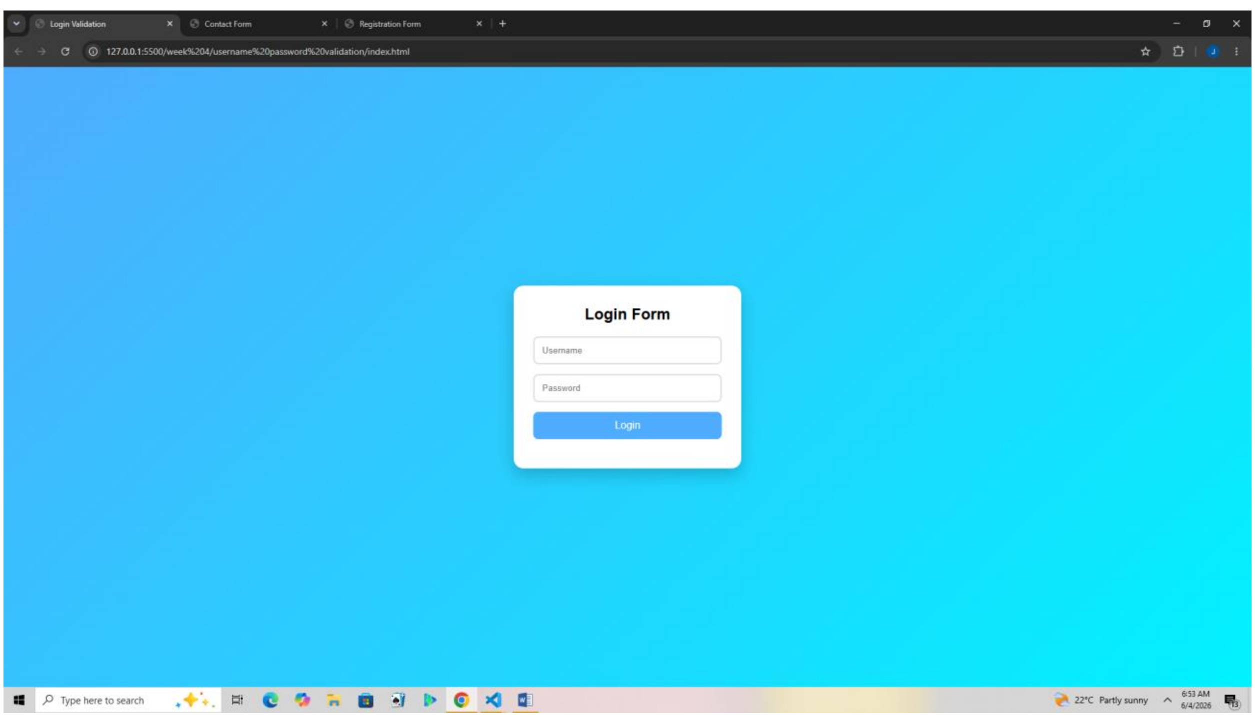


Fig 5: Contact form submission example

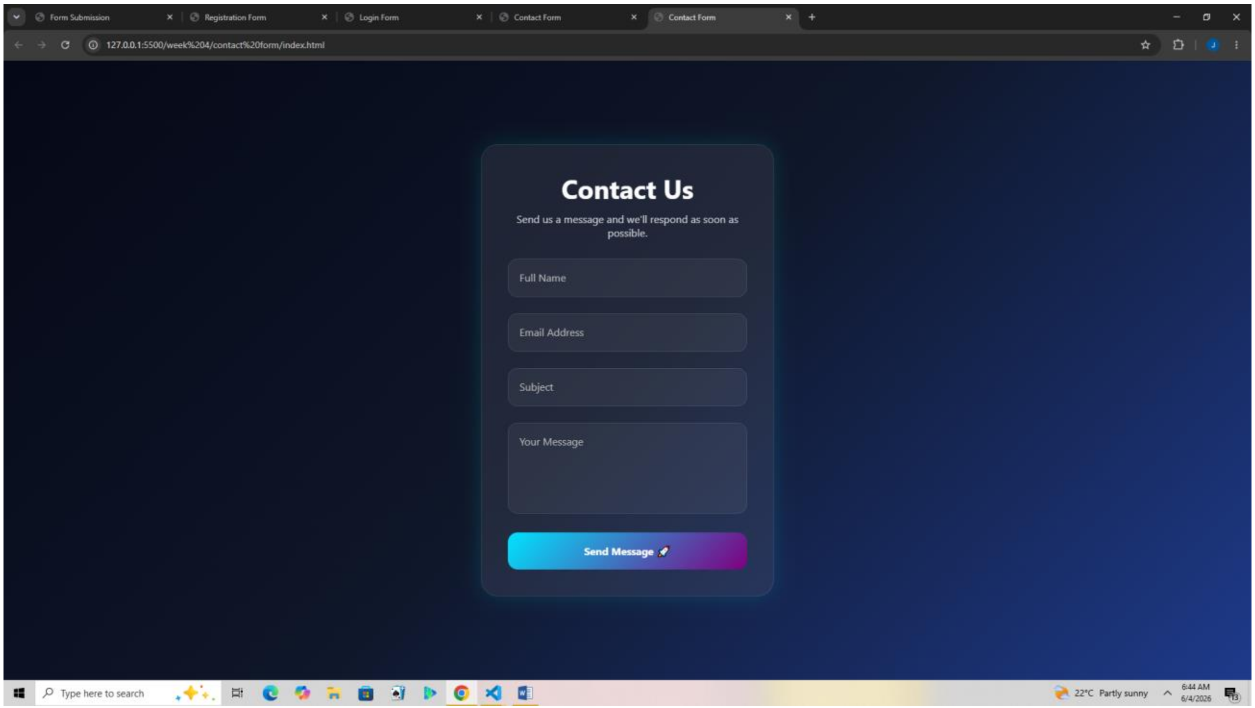


Fig 6: Basic login page with conditional access logic

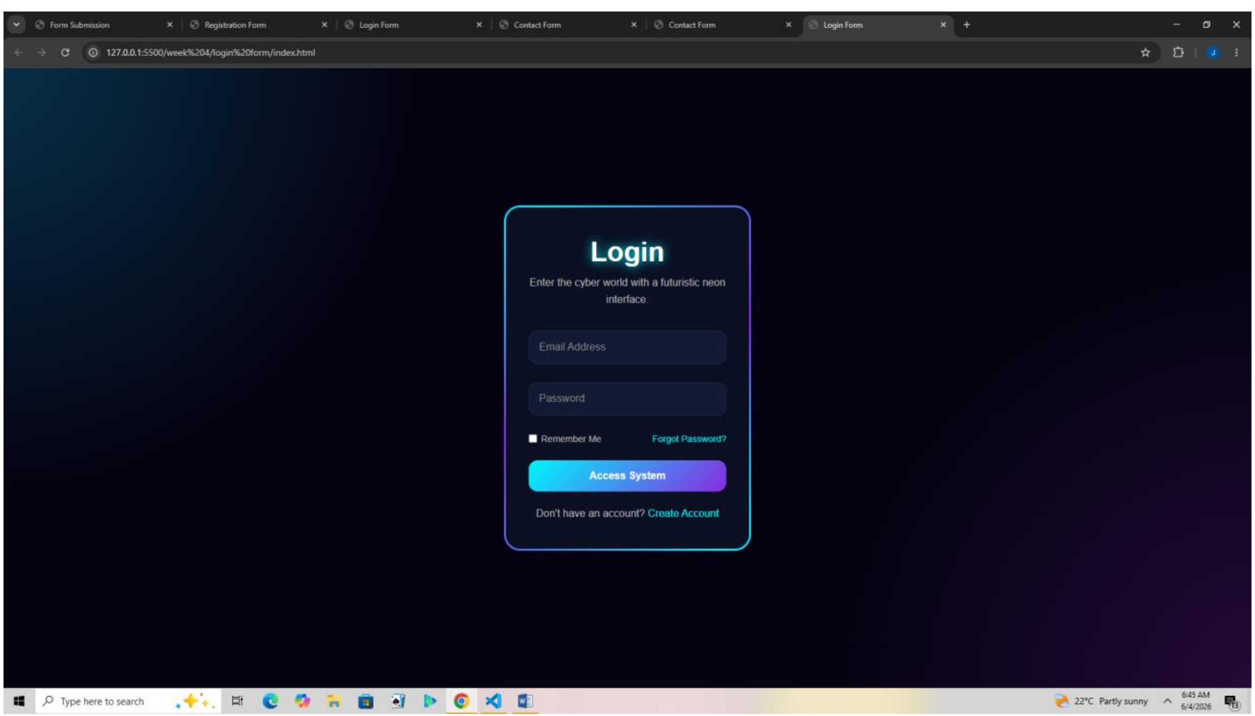
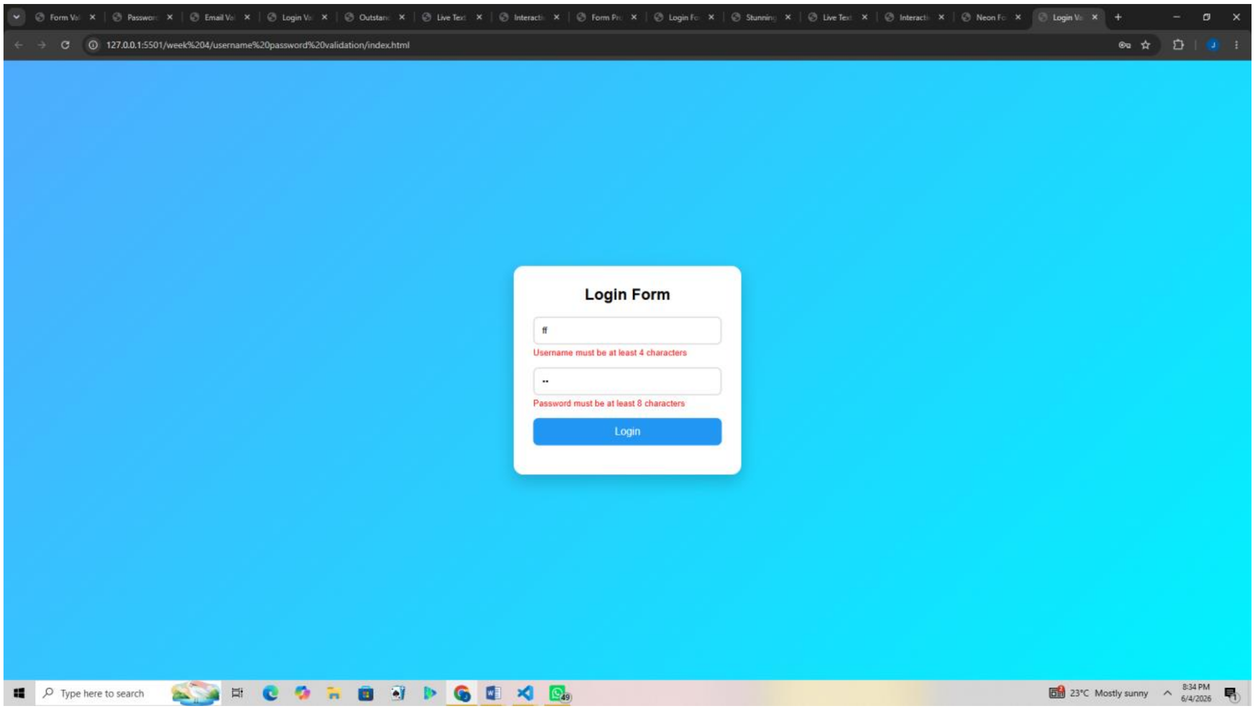


Fig 7: Username and password validation script ensuring secure login



REFLECTION

This week I learned how server-side programming processes user requests. Fig 1–3 show PHP handling form submissions and generating dynamic content. Fig 4–6 demonstrate registration, login, and contact forms integrated with backend logic. Fig 7–8 confirm authentication with username/password validation, while Fig 9 shows session handling for personalized user experiences. Organizing files into a backend structure improved workflow. The main challenge was ensuring secure handling of form data, which taught me the importance of using POST methods and validating inputs before granting access

WEEK 5:DATABASE

Fig1: phpMyAdmin showing the database successfully created.

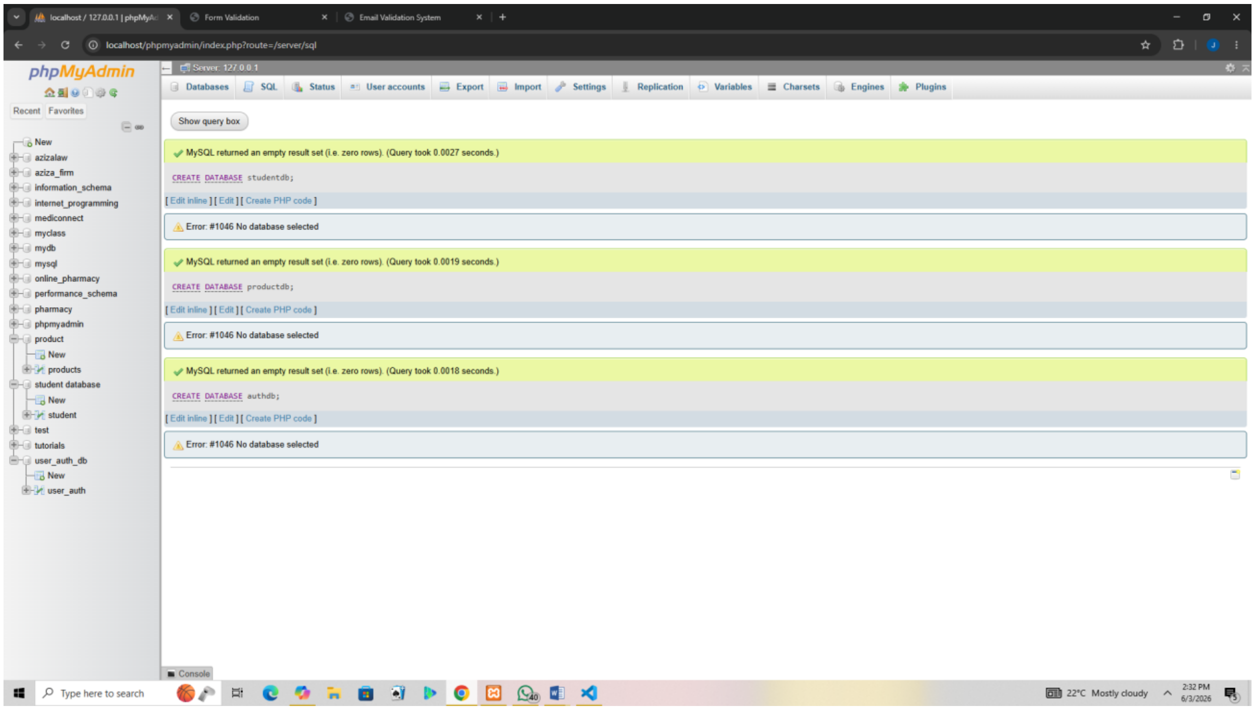


Fig 2: Table creation for users with ID, username, and password fields

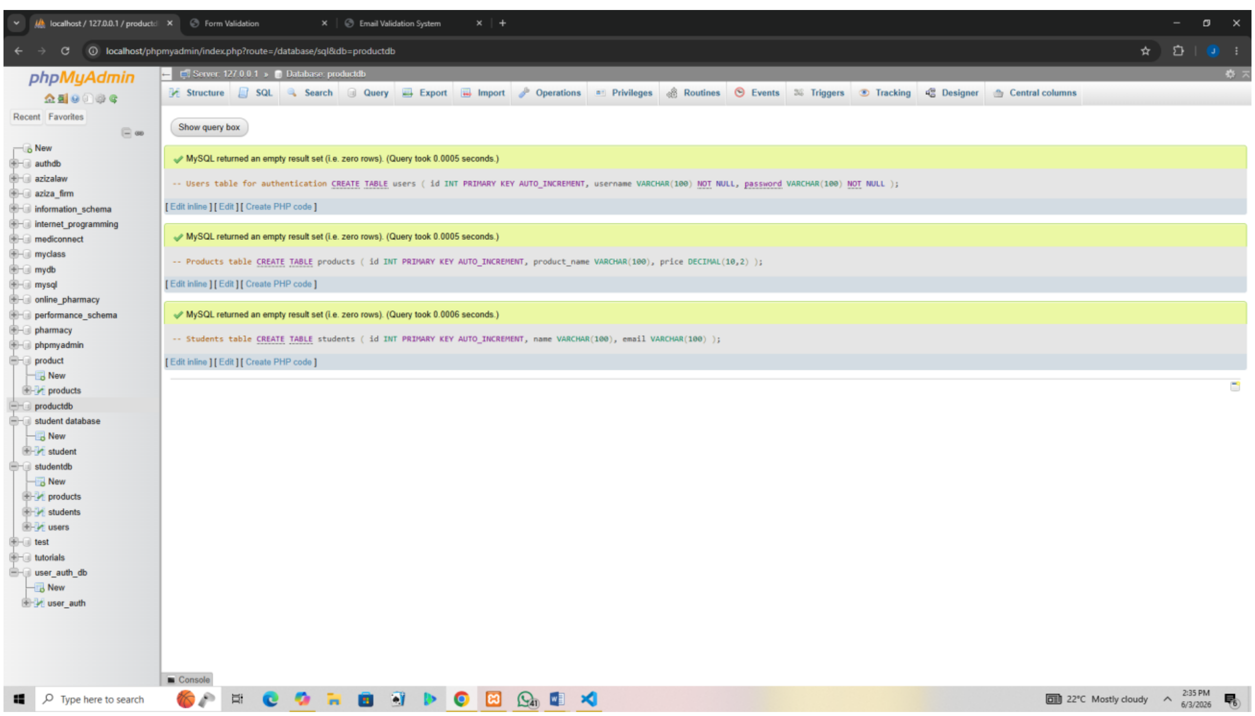


Fig 4: Insert query adding new student record

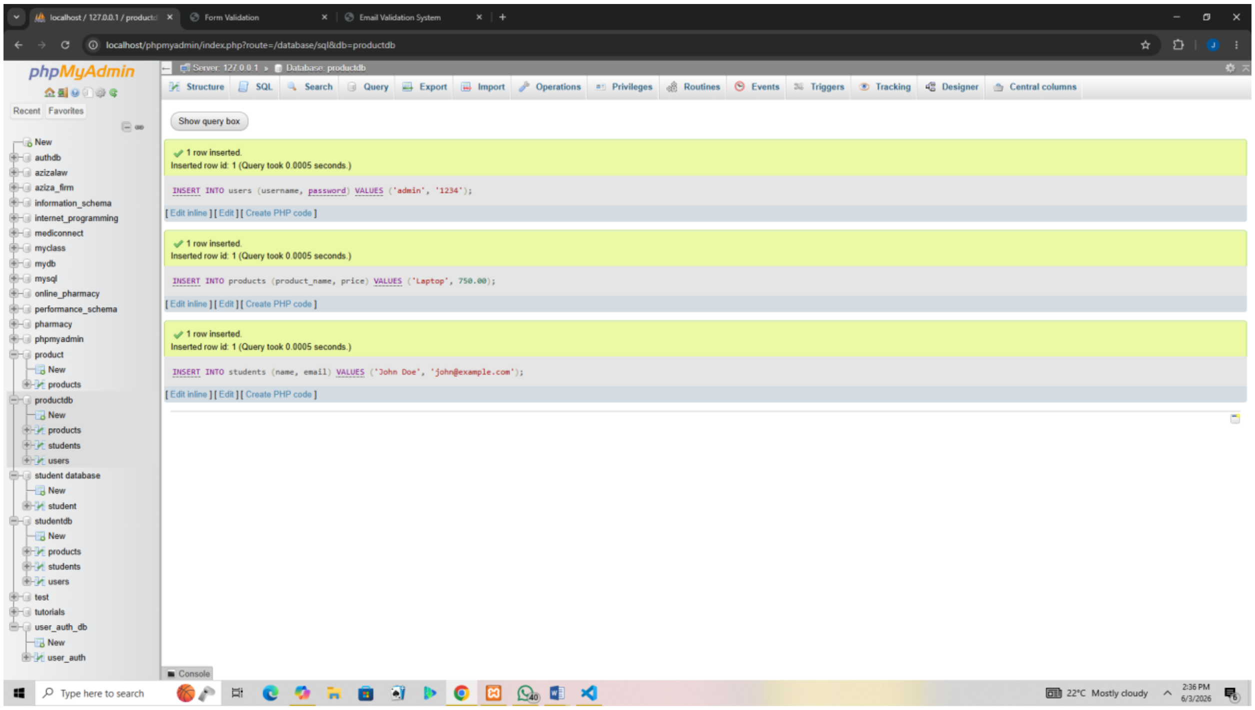
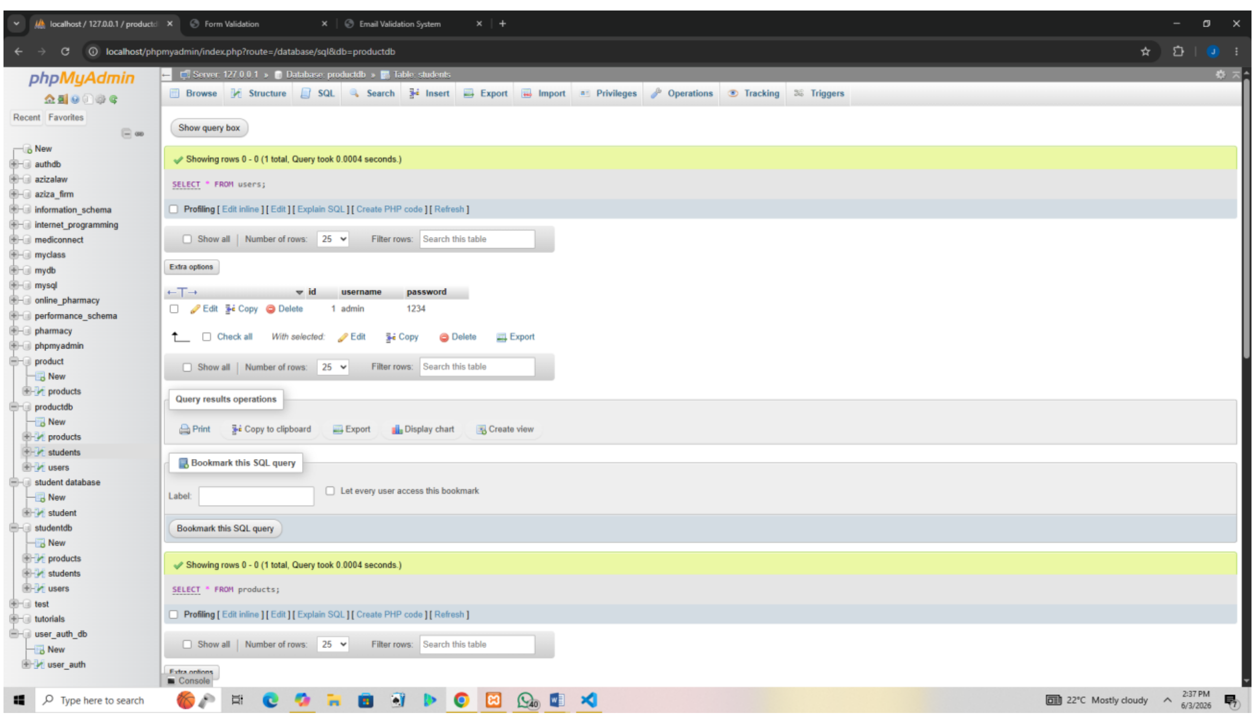


Fig 5: View records query displaying stored data



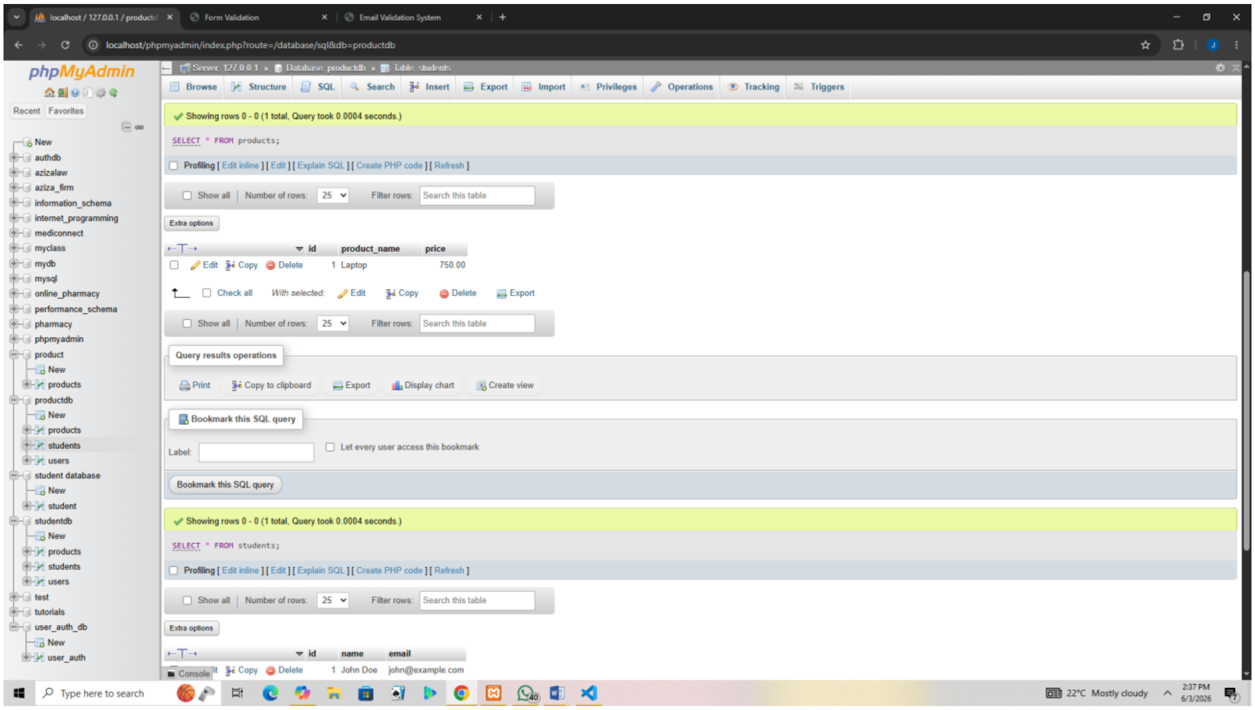


Fig 6: Edit query updating user information

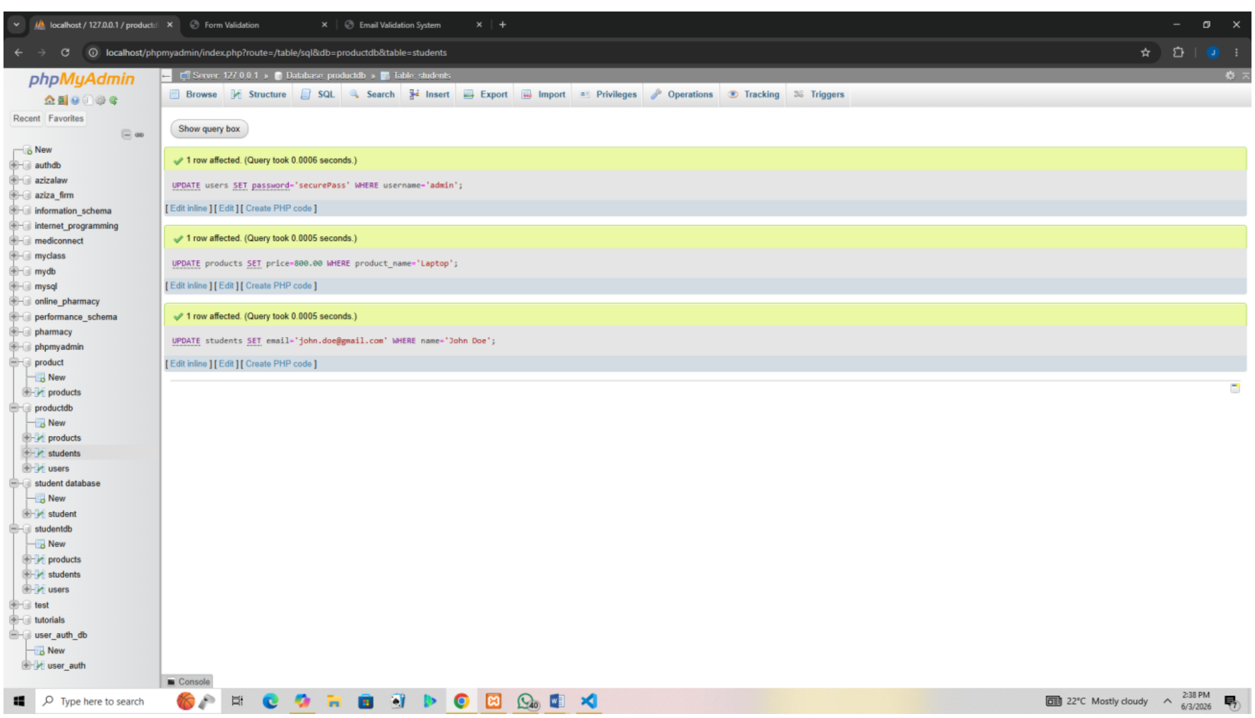


Fig 8: Delete query removing a record successfully(codes)

```
DELETE FROM users WHERE username='admin';

DELETE FROM products WHERE product_name='Laptop';

DELETE FROM students WHERE name='John Doe';
```

REFLEECTION

This week I learned how to build database-driven systems. Fig 1–3 show the creation of student, product, and authentication databases. Fig 4 demonstrates table setup for user credentials. CRUD operations (Fig 5–8) allowed me to add, view, edit, and delete records, proving full database functionality. These tasks taught me how backend systems manage persistent data and connect to frontend forms. The main challenge was ensuring queries executed correctly, which improved my understanding of SQL syntax and database integration with PHP.